

HTC Core: Integrated Procurement & Financial Tracking System

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CHAPTER 1

THE PROBLEM AND ITS BACKGROUND

Introduction

Heindrich Trading Corporation (HTC) is a Makati-based bulk commodity trader and a vital link in the alcohol industry. Despite its lean size, HTC secures massive molasses supply agreements, managing complex land and sea logistics directly from industrial sugar mills like BUSCO to beverage giants such as San Miguel, Tanduay, and Emperador. In 2026 alone, HTC handled 89,190.739 metric tons (MT) in deliveries (Heindrich Trading Corporation, 2026). However, operating on tight cash-flow margins with massive cargo volumes creates severe financial exposure. To fund upfront procurement and settle 50% logistics deposits, HTC relies heavily on short-term bank loans. Because client payments frequently stretch beyond standard 30-day windows, the company faces compounding bank interest that eats directly into profit margins.

The gravity of this exposure is highlighted in their company books as shown in Appendix B: despite an accepted 1% volume tolerance for transit tank fluctuations, HTC still incurred a staggering net loss of ₱2,343,480.55 (Heindrich Trading Corporation, 2026). This loss stems directly from three unmanaged areas: transit variances, operational complexities, and overdue loans. Specifically, client-specific leaks show losses scaling upwards from BDI (₱320,267.85) and Progreen (₱327,338.40), spiking through GSMI

(P439,866.0) and Emperador (P511,759.50), and peaking heavily with ADI at P744,248.80 (Heindrich Trading Corporation, 2026).

Presently, these variables including cash vouchers, bank loans, and barge logistics are managed via a fragmented, manual workflow of physical index cards and a single local Excel file. This manual dependency creates a dangerous blind spot: staff cannot manually compute daily compounding interest or cross-route volumetric shrinkage fast enough to catch margin erosion before it occurs. To mitigate these risks, HTC needs to transition to a centralized digital framework. By bypassing expensive physical hardware sensors, this study implements an intelligent, pure-software solution. Developing a dedicated platform featuring dynamic interest accrual tracking, asynchronous background transit volume calculations, and strict user permission boundaries will eliminate manual leaks, optimize capital loan lifecycles, and secure HTC's financial bottom line.

Statement of the Problem

Heindrich Trading Co. currently manages its complex bulk procurement, sales logging, and logistics tracking through manual, spreadsheet-reliant legacy workflows. This operational setup leads to several core problems:

1. Centralized Data Fragmentation and Cross-Generational Operational Friction:
Because HTC's lean team operates across contrasting generational methodologies ranging from localized spreadsheet consolidation by new hires to physical index-card tracking by senior personnel, transactional updates remain heavily isolated. This data fragmentation creates severe blind spots where real-time operational status updates cannot be shared concurrently across departments.
2. Cognitive Overstimulation and Risk Exposure in Manual Cash-Flow Reconciliation: The financial workflow demands a 50% upfront downpayment to logistics providers before receiving client settlements, requiring staff to manually cross-reference check payouts with complex multi-load logistics invoices. Managing this multi-tiered matching process through raw, number-heavy sheets creates high cognitive overstimulation for senior users, increasing the risk of delayed reconciliation and unmonitored profit margin erosion.
3. Passive Volumetric Loss Monitoring and Lack of Centralized Verification: Due to environmental factors and moisture drops during bulk commodity transport, the volumes loaded at sugar mills fluctuate before final delivery. Without a single-click verification interface, these critical transit discrepancies are reviewed

retrospectively on paper, making it difficult for the current personnel layout to efficiently monitor shrinkage and apply immediate financial adjustments.

We identified these operational challenges by conducting an extensive one-on-one interview with an Operations Staff member; please refer to **Appendix A** for the complete breakdown of the interview transcript

Project Objectives

General Objective

To develop and implement HTC Core, a secure, low-cognitive-overhead management information system engineered to transition Heindrich Trading Corporation's high-stakes, paper-based cash-flow workflows into an ergonomic web environment; the system consolidates the end-to-end transaction lifecycle while strictly adhering to the enterprise's custom operational specifications to eliminate cross-departmental data fragmentation.

Specific Objectives

1. To Implement a Method for Cross-Generational Data Aggregation and Cognitive Abstraction: To engineer a Django backend mapping layer that instantly consolidates decentralized spreadsheet rows and manual entry logs into a high-contrast dashboard, abstracting complex numerical grids into single-click visual status cards.

2. To Deploy a Multi-Tiered Financial Lifecycle Engine with a Dynamic Interest Accrual Method: To build a specialized accounting backend that binds active bank loans to physical cheques and 50% upfront logistics deposits; the system utilizes user-inputted interest rates and an adjustable aging calendar to run real-time interest clocks and visual timeline alerts that optimize capital repayment.

3. To Integrate an Asynchronous Volume Variance Tracker and an Immutable System Auditing Layer: To configure Celery background workers and Redis to automatically calculate volumetric shipping shrinkage against standard tolerances in return developing a predictive algorithm while establishing an unerasable database audit trail that logs and timestamps every financial state change and cheque modification for high-value fraud prevention.

Significance of the Study

The development and implementation of this cloud-based Management Information System are driven by a primary mandate: to modernize, secure, and streamline the core operational ecosystem of Heindrich Trading Co., specifically engineered to optimize business oversight and bring maximum convenience to its management and employees. By transforming fragmented, manual tracking routines into a unified, high-performance cloud environment, this study delivers profound strategic and practical value to its stakeholders:

Mr. Henry Chang: This project is structurally dedicated to providing total convenience, structural clarity, and administrative ease to corporate ownership. Recognizing that executive decision-making requires instant access to accurate data, this system eliminates the daily friction of auditing manual spreadsheets and chasing disjointed reports. It presents Mr. Henry Chang with a centralized, elegant administrative dashboard that consolidates complex logistics, procurement, and financial data into clear, real-time business reports. Furthermore, because the enterprise deals in life-changing amounts of money via physical cheques, the system specifically provides ownership with an Immutable System Audit Trail. This unalterable security layer allows management to monitor high-value check movements and data modifications with absolute security and fraud protection, granting total peace of mind over the company's trading lifecycle.

Operations and Invoicing Staff: The system significantly eases the daily operational burden on logistics and frontline documentation staff by replacing tedious manual coordination with structured automated workflows. Built-in database constraints prevent entry errors while providing a clean, high-contrast digital interface that simplifies complex data-entry workflows. Most importantly, rather than forcing personnel to calculate volumetric transit shrinkage manually on paper, the system utilizes asynchronous background scripts to automatically run volume variance calculations, instantly triggering visual alarms if cargo drops exceed standard tolerances. This guarantees immediate operational consistency and lowers execution risks.

Finance Personnel: This study introduces native financial intelligence that removes the heavy lifting from the company's fiscal workflows. Crucially, the system frees senior finance staff from manually tracking compounding bank loan interest on physical index cards. By deploying a dynamic interest accrual engine, the platform provides automated payment-to-expense matching scripts that seamlessly tie sales invoices to cash vouchers, active bank loans, and 50% upfront logistics deposits. This protects the ledger from human arithmetic errors and accelerates settlement timelines by visually alerting staff precisely when to close active loans to prevent margin erosion.

The Industry and Small-Scale Commodity Distributors: This study establishes an empirical precedent proving that independent trading companies can attain high-level enterprise resource planning (ERP) capabilities without taking on the capital-intensive infrastructure dependencies (e.g., dedicated hardware arrays, private blockchain

architectures) heavily criticized in modern literature. It offers a low-cost, software-driven blueprint for organizational modernization.

Future Researchers and Academics: This research contributes a reliable academic framework for the design and deployment of secure, high-concurrency cloud systems utilizing a decoupled multi-tier stack (NGINX, Gunicorn, Django, and PostgreSQL hosted on AWS EC2). It serves as a thorough reference for future scholars specializing in secure application engineering and agile development methodologies within localized commodity trading markets.

Scope and Limitations

Scope

The project focuses on building a dedicated web platform to manage the core local trading, logistics, and internal reconciliation operations of Heindrich Trading Co. The system scope is defined by the following features:

1. Core Modules: Contract and Purchase Order tracking, multi-variable logistics ledger (barge and tracking service records), dynamic input for transit quantity variances (delivered vs. received volume), and dual-invoice cross-referencing (supplier invoice numbers linked to internal Cash Voucher and Sales Invoice numbers).

2. Automation Logic: Real-time software alerts driven by database constraints to flag contract fulfillment progress, missing expense associations, and significant volume discrepancies exceeding standard tolerance thresholds.
3. Access and Control Security: A secure login framework with multi-role permissions that splits views and capabilities between administrative users and standard employees.
4. Deployment Architecture: A cloud-hosted web system that runs efficiently across modern desktop web browsers, ensuring remote accessibility and collaborative data updating without local server maintenance.

Limitations

To maintain project focus and stay within budget, the system operates under the following boundaries:

1. Hardware Exclusions: The platform uses pure software and database logic to log transaction records and track logistical costs. It does not integrate with external hardware tracking arrays, such as RFID chips, barcode scanners, or automated GPS tracking units hosted on physical barges.

2. Financial and Accounting Exclusions: The system focuses strictly on operational procurement, sales logging, and transactional payment-to-expense matching. It does not provide full corporate accounting capabilities, general ledger generation, balance sheet balancing, or automated tax localization modules.
3. Financial Confidentiality and Structural Exclusions: Heindrich Trading Co. has reserved the right to keep the depth of their financial records, statements, and corporate dealings strictly confidential due to the substantial monetary amounts involved and sensitive trade secrets. Consequently, the system focuses solely on internal procurement and sales logging. It explicitly excludes profit-and-loss auditing, tax localization modules, or automated multi-currency financial ledger systems.
4. E-Commerce Boundaries: The interface is built strictly for internal business administration, management tracking, and back-office billing log synchronization. It does not include a public-facing customer storefront, broker interaction portals, or online customer payment gateways.
5. SRA Regulatory Automation Limits: While the platform stores sourcing data types (local vs. imported molasses metrics), it will not dynamically automate or enforce complex Sugar Regulatory Administration compliance rules (such as the 3:1 local-to-import rule or customs clearance document tracking) within the system architecture. These remain external compliance workflows.

6. Tank Fluctuations: Environmental volume shifts are a fixed physical constraint.
We absorb the 1% transit tolerance as per the company's protocol.

Definition of Terms

The following definitions of the terminologies used by the researchers are stated below to better understand the study:

Asynchronous Background Worker – A software architecture pattern implemented via Celery that executes intensive data processing tasks—such as transit shrinkage calculations—independently of the main user interface thread to prevent application lag.

Barge Logistics Ledger – A specialized tracking module within the database designed to log maritime transport variables, including vessel identification, loading/unloading timestamps, carrier assignments, and associated transit fees.

Brix Penalty – A commodity-specific financial adjustment applied to molasses transactions where the unit price is docked or prorated due to density, moisture, or quality degradation detected during transport.

Bulk Commodity Trading – The commercial buying and selling of large quantities of raw, unpackaged goods (specifically molasses in this study) requiring specialized multi-modal transport and high-volume contract management.

Business Rules – Programmatic constraints hard-coded into the software logic to

automate and enforce company policies, such as blocking duplicate entries or throwing alerts when delivery discrepancies exceed accepted tolerance limits.

Capital Loan Tracking – The digital logging and monitoring of short-term financing loans and associated physical paper cheques secured by the enterprise to fund upfront procurement and logistics operations.

Cash Voucher Number – A unique internal alphanumeric identifier generated by the system to authorize and track cash disbursements distributed upfront to logistics providers and sugar mills.

Dynamic Interest Accrual Engine – A custom accounting database method that calculates daily interest liabilities on open bank capital loans using user-inputted percentage rates and variable chronological tracking thresholds.

Immutable System Audit Trail – An unalterable, append-only security ledger within the database architecture that permanently timestamps and records user identities alongside all financial state changes and cheque modifications to prevent high-value data fraud.

In-Memory Data Structure Store – A high-performance database caching framework implemented via Redis that acts as a message broker to manage tasks handled by asynchronous background workers.

Invoicing Staff – The user role assigned to employees responsible for inputting outward-facing billing logs, generating sales invoices, and cross-referencing supplier documents.

Iterative Agile Methodology – A software development approach where projects are built in small, manageable cycles called iterations, emphasizing continuous stakeholder feedback, adaptability, and progressive system improvements.

Legacy Workflow – The traditional, manual, and paper-reliant business practices currently utilized by Heindrich Trading Co., which heavily depend on physical index cards and a single, local Excel file.

Multi-Tier Role Permissions – A software security framework that restricts or grants system features, access points, and dashboard views depending on a user's authenticated organizational role (Management, Finance Officer, Invoicing Staff, Operations Staff).

Operations Staff – The user role designated for personnel overseeing field logistics, contract entry, mill dispatches, barge movements, and transit volume parameters.

Payment-to-Expense Matching Logic – Automated ledger scripts within the software that allow the finance department to group incoming customer payments directly against the specific cluster of expenses (sourcing, tracking fees, barge costs) incurred by that exact transaction.

Purchase Order – The digital data entry record generated in the system upon management's communication channel (Viber) approval, confirming the pricing, volumes, and terms agreed upon for a shipment.

Predictive Algorithm - A software-driven computational process embedded in the backend system that evaluates transit discrepancies against a fixed 1% volumetric tolerance threshold to automatically adjust billing and prevent financial loss.

Sales Invoice Number – The unique, system-generated digital identifier mapped to an outward transaction billing log to track client accounts receivable.

Single-Click Execution Model – A simplified user interface design approach optimized for non-technical users, abstracting complex web workflows and database transformations behind intuitive, high-contrast status cards to minimize cognitive overstimulation.

Transit Quantity Variance – The calculated difference between the volume of molasses originally loaded at the sugar mill versus the volume recorded upon delivery at the final client location caused by environmental factors during transit.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter establishes the technical, operational, and empirical foundations for the proposed management information system at Heindrich Trading Co. Centralizing and securing data pipelines is now imperative to eliminate fragmented legacy environments and sustain organizational throughput. Specifically, this review explores four unified themes:

1. Web-Based and Cloud-Based Supply Chain Architectures, which outline the structural blueprints for distributed system deployment;
2. Management Control Frameworks and System Integrity Validations, defining programmatic control layers and multi-tier security boundaries.
3. Telemetric Networks, Distributed Ledgers, and Real-Time Asset Tracking, evaluating high-performance tracking and database logic paradigms.
4. Systematization and Meta-Analyses of Digital Transformation Trends, providing macro-level validation for systemic digitization.

By evaluating these models, this review isolates the practical constraints of capital-intensive frameworks, establishing an empirical justification for a software-driven, cost-effective web platform designed to optimize multi-variable contract tracking, contract-to-expense matching, and collaborative transaction updating.

Web-Based and Cloud-Based Supply Chain Architectures

Transitioning from physical reporting to scalable web and cloud infrastructure is critical for fluid workflows. Yu and Zhou (2020) demonstrated that multi-tier supply chain platforms using distributed relational databases effectively bridge information silos among suppliers, manufacturers, and consumers. Node optimization and network synchronization protocols further accelerate data processing speeds across these distributed database clusters (Wang, 2021). Consequently, migrating organizational data to centralized cloud repositories significantly optimizes cross-departmental coordination and overall operational performance (Dong & Salwana, 2022). While these architectures provide robust enterprise blueprints, they focus on macro-level operations with extensive capital backing, omitting localized workflows for independent commodity distributors who must handle rapid sourcing shifts due to volatile supplier inventories. The software platform for Heindrich Trading Co. addresses this gap by adapting multi-tier concepts into a lightweight, collaborative web portal, replacing a manual, single-user legacy setup with real-time cloud data entry and contract tracking.

Management Control Frameworks and System Integrity Validations

To protect these pipelines, transparent data monitoring configurations, immutable transaction logs, and rigid role validations mitigate fraud and entry blind spots (Leong et al., 2024). In platforms managing high-value assets and physical cheque disbursements, embedding automated, unalterable database audit trails directly reduces internal security

risks. Embedding this real-time data validation transforms software from a passive archive into an active risk-mitigation tool (Fatima Ezzahra et al., 2025). However, existing literature remains heavily conceptual or restricted to large-scale corporate manufacturing. The proposed system translates these compliance principles into a localized environment for Heindrich Trading Co. by hard-coding distinct user privilege boundaries and establishing an immutable system auditing layer to securely track all financial state changes and late-arriving customer payments.

Telemetric Networks, Distributed Ledgers, and Real-Time Asset Tracking

This is supported by Pethe et al. (2024), who evaluated telemetric asset-tracking networks using edge-to-cloud device gateways. Their framework successfully deployed automated notification scripts and threshold indicators to minimize logistical downtime. However, these existing frameworks depend on capital-intensive infrastructures, such as external RFID/GPS sensor hardware arrays or complex blockchain networks, making them inaccessible for small trading businesses. The proposed system addresses this limitation by deploying asynchronous background scripts specifically utilizing Celery task queues and Redis message brokers as a low-cost, pure-software alternative. Instead of using expensive external sensors, these background workers execute volume variance tracking natively through database constraints, instantly flagging transit discrepancies without slowing down the user interface.

Systematization and Meta-Analyses of Digital Transformation Trends

Macro-level academic evaluations establish clear baselines for digital system integration within the logistics sector. A meta-analysis of 1,264 academic papers mathematically validated that integrating IT-based supply chain management platforms directly enhances global transparency, inter-organizational communication, and sustainable business operations (Chiang et al., 2021). Structured coordination software consistently prevents operational friction, eliminates administrative backlogs, and reduces delivery delays across logistics channels (Reznik et al., 2022). Because these systematic reviews focus on global, macro-enterprise trends, they explicitly conclude by highlighting a critical need for localized empirical studies to test custom software configurations in real-world settings. Developing a custom management information system for Heindrich Trading Co. directly satisfies this requirement, serving as a localized empirical test case that applies these broad, systemic principles to optimize a small-scale commodity trading enterprise.

Table 2.1 – References

Author & Date Published	Description	Technology Features Used	Gaps in Research
Wang (2021)	The study developed an intelligent data	- Horizontal system integration	This paper didn't consider what

Author & Date Published	Description	Technology Features Used	Gaps in Research
	framework mapping information flow patterns across distributed enterprise networks. It explored the ways in which agile network configurations resolve low informational awareness, loose business compliance, and fragmented logistical communication trails. Results have revealed that there is an improvement in data processing speeds	- Localized sub-database clusters - Node optimization algorithms - Network time synchronization protocols - Distributed asset monitoring paths - Graphically displayed layout layers	happens with these distributed nodes when they control certain retail activities, are involved in localized invoicing, or manually track goods in the warehouse within an independent distributor organization. The Heindrich Trading Co. software implementation addresses this apparent “gap” in

Author & Date Published	Description	Technology Features Used	Gaps in Research
	and a marked optimization in distributed cluster synchronization when deploying algorithmic node mapping.		conception by conducting a gap analysis of the manual procedure that is being repeated in the business.
Bhanu & Krithika (2025)	Proposed a structural framework verifying how information reliability scales organizational throughput via transparency metrics. Proved that content, scope, and data comprehensiveness account for a 75%	- Fuzzy Analytic Network Process (ANP) - Data Envelopment Analysis (DEA) - Dimensional transparency indexing (timeliness, regularity, scope)	Heavily dependent on macro-level analytics matrices. It bypasses low-level relational validation architectures required by independent commodity traders to track

Author & Date Published	Description	Technology Features Used	Gaps in Research
	direct impact on supply chain transparency factors.		cross-referenced logistics expenses and real-time shipment discrepancies.
Mirbagheri et al. (2023)	The study engineered a data-driven Decision Support System (DSS) utilizing a centralized data warehouse framework to maximize supply chain transparency. Mirbagheri's experimental findings demonstrated that unifying isolated legacy operations into	- Data-Driven Decision Support Systems (DSS) - Centralized data warehouse architectures - Extract, Transform, Load (ETL) tools - Real-time business intelligence dashboards - Historical trend analysis algorithms - Data profiling and cleansing protocols	A distinct limitation of this framework is its reliance on pre-existing enterprise data structures, bypassing the low-level relational database schemas and user-access configurations necessary for

Author & Date Published	Description	Technology Features Used	Gaps in Research
	structured storage clusters empowers managers to monitor real-time stock balances and detect seasonal distribution trends.		independent retail merchants.
Leong et al. (2024)	Evaluated how integrating strict business ethics and transparent supplier criteria into corporate operational workflows builds marketplace trust, reduces legal risk, and optimizes logistics sustainability.	- Immutable transaction recording layers - Multi-tier user privilege boundaries - Anti-corruption data architectures - Automated transparency dashboards - Audit trail mapping frameworks	Focuses heavily on high-level corporate policy guidelines, supplier screening strategies, and global environmental metrics. It provides no technical blueprints for

Author & Date Published	Description	Technology Features Used	Gaps in Research
			lightweight web database constraints that address manual data-handling friction within small administrative teams.
Fatima Ezzahra et al. (2025)	Investigated the evolution of digital management systems, proving that transitioning away from traditional, retrospective data filing into active validation layers converts software into	- Real-time operational data pipelines - End-to-end operational visibility modules - Dynamic performance metrics	Highly conceptual and enterprise-centric in scope. It does not detail the practical, browser-accessibl e database schemas or single-click user

Author & Date Published	Description	Technology Features Used	Gaps in Research
	a real-time risk mitigation tool.		flows needed by small trading firms to track fluid commodity shipments across third-party logistics networks.
Yu, X., & Zhou, Q. (2020)	Focuses on an ASP.NET-based web application designed to integrate macro-level supply chain layers (suppliers, manufacturers, distributors, and end customers) for	- Browser/Server (B/S) and Client/Server (C/S) hybrid setups - Three-tier architecture (browser, application server, database server) - Distributed database support (MySQL, Oracle, SQL Server)	Focuses strictly on macro-enterprise architectures. It omits implementation data, financial cost barriers, and mobile-enabled deployment strategies for

Author & Date Published	Description	Technology Features Used	Gaps in Research
	real-time tracking and decision support.	- Electronic Data Interchange (EDI) and ERP integration - GPS/GIS tracking and barcode data collection	small, independent distributors.
Dong, X., & Salwana, E. (2022)	Analyzes the quantitative impact of migrating human resource and supply chain management data to centralized cloud repositories to optimize cross-departmental operations.	- Cloud-Based Supply Chain Management (CBSCM) - Cloud-Based Human Resource Management (CBHR) - Centralized database coordination paths - Cross-departmental digital workspaces	Assumes pre-existing corporate digital infrastructure. It fails to map specific database schemas or data entry transition models needed to convert manual paper records into digital formats.

Author & Date Published	Description	Technology Features Used	Gaps in Research
Hijjawi, H., et al. (2023)	Investigates how digital supply networks act as a mediator to optimize operational performance and communication structures when paired with lean management practices.	- Structural Equation Modelling (SEM) - SmartPLS analysis software - Digital supply chain integration frameworks - Structural data communication dimensions	Evaluated exclusively within large-scale manufacturing firms. It fails to address localized inventory tracking, direct over-the-counter billing, or manual bookkeeping challenges common to small retail/wholesale businesses.
Maheshwari, S., Gautam,	Identifies security risks and information delays caused by	- Big Data Analytics (BDA) frameworks	Requires advanced, high-overhead big

Author & Date Published	Description	Technology Features Used	Gaps in Research
P., & Jaggi, C. K. (2021)	traditional siloed data storage, proposing real-time analytical dashboards for proactive decision-making.	- Real-time analytical tracking tools - Automated data visualizations and dashboards - Predictive and descriptive diagnostic algorithms	data engineering infrastructures. It completely skips basic web application configurations needed by standalone vendors to track stock without a dedicated data team.
Dashtizadeh, M., Meskaran, F., & Tan, D. (2022)	Details a secure, permissioned cryptographic ledger framework designed to record transaction histories and track	- Cryptographic hash functions and asymmetric-key encryption - Smart contracts via Solidity script execution	Centered heavily on specialized pharmaceutical supply chains (vaccine verification). It

Author & Date Published	Description	Technology Features Used	Gaps in Research
	items across distribution networks.	- Decentralized ledger systems - React, HTML, CSS, and JavaScript front-end UI - Node.js server-side scripting	does not provide cost-benefit metrics or interface usability tests for low-budget, standard trading environments.
Nofandi, F., Janaputra, M. R. B., & Pratama, R. A. (2025)	Engineered a web-based decentralized supply network configuration designed to combat institutional mistrust, transactional misinformation, and delayed operational reporting.	- Prototyping methodology - Distributed tracking ledgers - Real-time multi-actor sync (suppliers, processors, distributors)	Relies on decentralized blockchain ledger systems which impose prohibitive capital and configuration overheads for small-scale operations. It

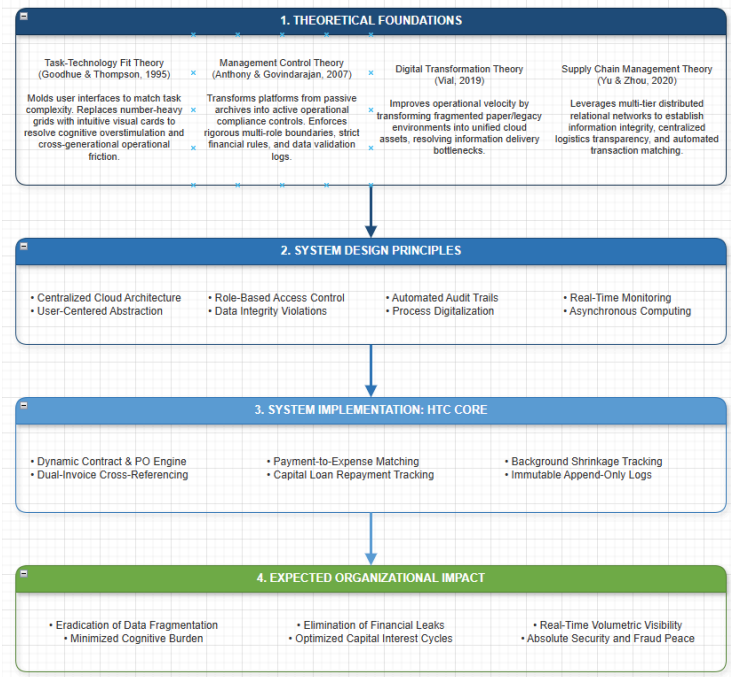
Author & Date Published	Description	Technology Features Used	Gaps in Research
			completely omits a simplified, software-driven data workspace for contract-to-expense matching
Reznik, N., Rudenko, S., & Pylypchuk, K. (2022)	Maps out the conceptual foundations of integrated logistics systems, proving that coordinated digital networks prevent operational friction and reduce delivery delays.	- Integrated supply chain management frameworks - Process coordination modeling - Systematization and comparative analysis techniques	Entirely limited to theoretical and definitions-based analysis. It provides no actual system code, database frameworks, or empirical tests in a live business setting.

Author & Date Published	Description	Technology Features Used	Gaps in Research
Smerichevsk a, S., Prodanova, L., & Yakushev, O. (2024)	Contrasts static document digitization with active automated circulation, demonstrating that automating operational files reduces human data-entry mistakes.	- Electronic document management networks - Internet of Things (IoT) warehouse tracking - Cloud infrastructure hosting (AWS)Wi-Fi and Bluetooth communication protocols	Focuses on capital-intensive automation equipment (robotics, AI, complex IoT). It omits technical blueprints for a standalone web app that merges daily point-of-sale over-the-counter billing with basic inventory counts.
Pethe, S., et al. (2024)	Evaluates telemetric architectures that use automated threshold	- Real-time asset-tracking telemetric networks	Relies heavily on external physical hardware

Author & Date Published	Description	Technology Features Used	Gaps in Research
	triggers and warning systems to minimize warehouse stockouts and administrative downtime.	- Edge-to-cloud device gateways - Automated stock-level reorder notification scripts - Diagnostic analytics dashboards and relational data storage	deployment (RFID tags, GPS nodes). This introduces cost and setup barriers for small businesses looking for a purely software-driven database solution.
Chiang, C.-T., Kou, T.-C., & Koo, T.-L. (2021)	Evaluates historical IT-based logistics trends across 1,264 academic papers to outline how digital platforms improve transaction transparency and	- Keyword analysis frameworks - Social network analysis tools - Sustainable Supply Chain Management (SSCM) tracking metrics	As a meta-analysis, it provides only generalized macro-level policy suggestions. It specifically calls for localized

Author & Date Published	Description	Technology Features Used	Gaps in Research
	inter-organizational communication.		empirical studies to test custom web applications in small-scale business environments.

Theoretical Framework



Synthesis

The reviewed literature establishes that migrating from paper-based, manual operations to cloud-hosted relational database environments is crucial for optimizing business performance, reducing administrative error, and mitigating operational risks (Yu & Zhou, 2020; Dong & Salwana, 2022; Fatima Ezzahra et al., 2025). Existing frameworks confirm that embedding algorithmic validation checks and automated threshold tracking into software workflows consistently eliminates transaction visibility gaps and delivery backlogs. (Fu et al., 2022; Reznik et al., 2022; Pethe et al., 2024). However, a distinct gap exists in current research: existing models focus primarily on macro-enterprise operations with large capital backing, or rely on expensive hardware infrastructure like RFID, GPS, and blockchain networks (Maheshwari et al., 2021; Dashtizadeh et al., 2022). These frameworks omit low-overhead relational schemas tailored for independent commodity distributors who must handle specialized needs like multi-modal barge logistics, transit-driven volume-to-density fluctuations, and complex payment-to-expense voucher matching (Yu & Zhou, 2020; Pethe et al., 2024). Furthermore, corporate platforms often overlook the usability constraints of non-technical administrative staff who require single-click user flows. Our custom system bridges these academic gaps by bypassing expensive physical tracking hardware. Instead, it embeds cross-departmental data abstraction, dynamic interest-accrual modeling, and unalterable system audit trails directly into pure software logic, serving as a localized empirical test case for independent, high-stakes distributors.

CHAPTER 3

METHODOLOGY

This chapter describes how the proposed cloud-based management information system for Heindrich Trading Co. will be planned, built, and validated. Based on stakeholder interviews, the company's core need is to replace manual Excel-based molasses trading records with a centralized web platform that supports contract and purchase order tracking, delivery monitoring (including delivered versus received quantities), barge and tracking service costs, sales invoice and cash voucher references, per-customer dashboards, and payment–expense matching for finance staff. The methodology aligns with the objectives and scope in Chapters 1 and 2: a cost-effective, browser-based system with multi-tier user access, without RFID/IoT hardware, public e-commerce, or full corporate accounting. HTC (Heindrich Trading Co. internal entity referenced in operations) is used in records for storage and related transactions; the proposed system serves the same organization.

This chapter covers the development model, requirement analysis, hardware and software requirements, target users and environment, data gathering methods, and initial system design. **Descriptive research** was utilized to gather granular information regarding the company's existing workflows, transaction recording practices, inventory monitoring procedures, and financial tracking activities. While the core framework focus

remains developmental, descriptive research functions supplementarily to capture exact, real-world operational parameters that cannot be extracted from raw data models alone.

To achieve this, the researchers conducted structured and semi-structured qualitative interviews with key operations and finance employees within Heindrich Trading Co. Engaging directly with the staff members handling day-to-day data entry and fiscal reconciliations allowed the researchers to document exact pain points, identify where manual spreadsheet dependencies fail, and map empirical user requirements. The insights gathered from these personnel directly informed the relational database constraints and business logic rules necessary to establish an accurate operational framework.



The study adopts the Agile Software Development Model in designing and developing the proposed cloud-based Management Procurement System. Agile was selected because it allows continuous collaboration with stakeholders and supports incremental improvements throughout the development process.

The Agile approach enables the researchers to divide the project into smaller and manageable phases called sprints. Each sprint produces a functional system component that can be reviewed and evaluated by stakeholders. Feedback obtained from each iteration is incorporated into succeeding development cycles, ensuring that the final system effectively addresses the company's operational requirements.

The Agile methodology was preferred over the traditional Waterfall approach because business requirements may change during development, particularly in transaction processing, delivery monitoring, and financial tracking activities. Agile provides flexibility, faster feedback cycles, and improved adaptability to user needs (Chiang et al., 2021; Nofandi et al., 2025).

The Agile Development Cycle emphasizes continuous improvement through repeated iterations. Each cycle allows stakeholders to evaluate system functionality and recommend enhancements before proceeding to the next development stage. This iterative process helps ensure the successful development of a system that meets organizational requirement

Figure 3.1 – Input-Process-Output (IPO) Diagram

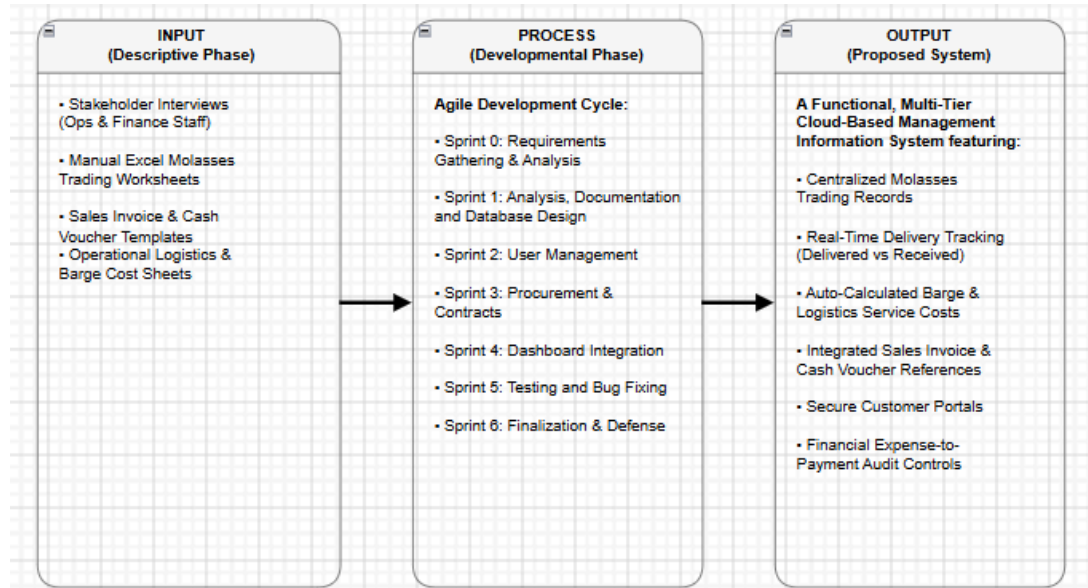
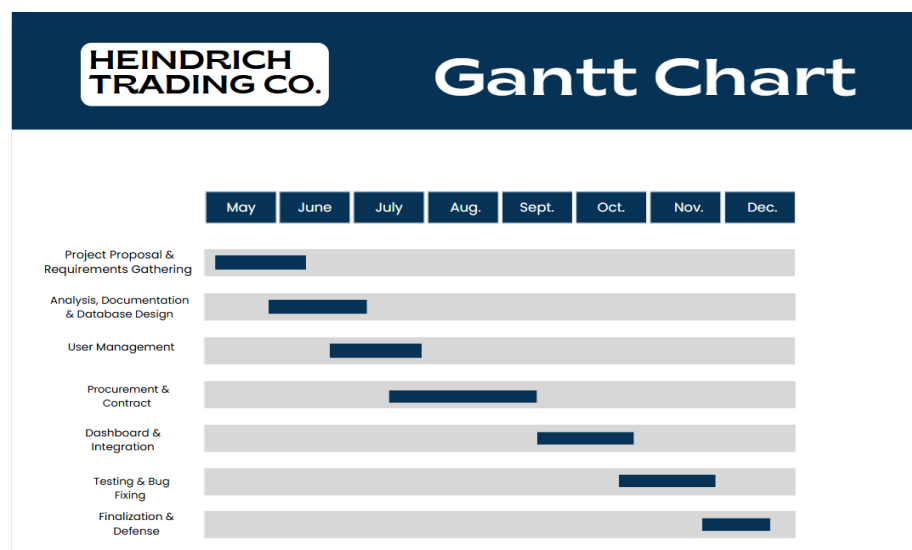


Table 3.1 – Sprint Plan

Sprint	Deliverable
Sprint 0	Requirements gathering and analysis
Sprint 1	Analysis, Documentation and Database Design
Sprint 2	User Management
Sprint 3	Procurement & Contracts
Sprint 4	Dashboard Integration

Sprint 5	Testing and Bug Fixing
Sprint 6	Finalization & Defense

Figure 3.2 – Gantt Chart



The proposed project follows an eight-month timeline spanning from May to December, transitioning systematically from descriptive research to iterative system development using the Agile framework. The timeline begins with a two-month inception and architectural phase (May–June) dedicated to planning, stakeholder interviews, requirement documentation, and database modeling. This is followed by a four-month core development period (June–October) executing Sprints 1 through 5, which systematically constructs user role management, the core molasses procurement and

contract engine, logistics and barge cost calculators, and integrated reporting dashboards. The timeline concludes with a two-month quality assurance and deployment phase (October–December) covering Sprint 6 system testing, bug fixing, official User Acceptance Testing (UAT) sign-off with Heindrich Trading Co. personnel, and final oral defense preparation.

Figure 3.3 – Work Breakdown Structure

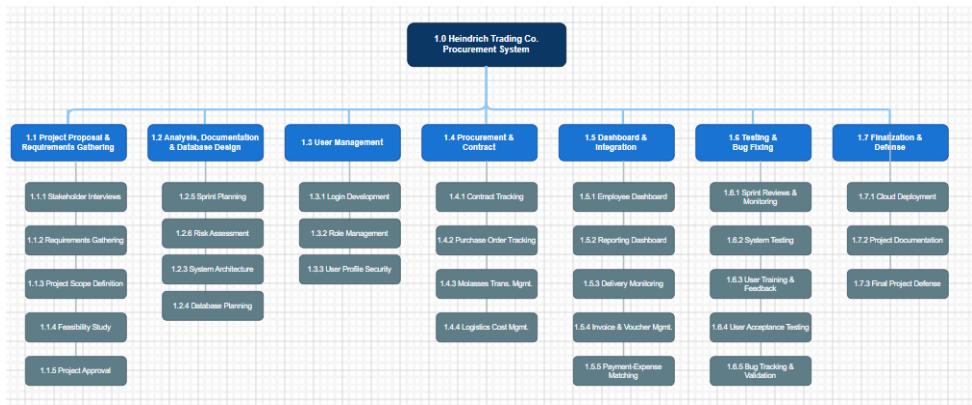


Table 3.2 – Team Roles and Responsibilities

Member	Role	Responsibilities
Bernardino, Denielle	UI/UX Designer	Owns the end-to-end user interface and experience design; creates wireframes, prototypes,

		and visual assets.
Cahindi, Joseph Francis	Full-Stack Engineer	Collaborates with the team leader on technical architecture; drives both frontend and backend code implementation.
Maranda, Sam	Research Lead	Manages project documentation, academic writing, and research papers.
Santos, Kurt Zildjian	Project Leader & Backend Engineer	Oversees project milestones, finalizes deliverables, and manages database architecture and software.

Constraints

Technical Constraints

Internet Connectivity

The proposed system is a cloud-hosted web application that requires a stable internet connection for real-time data synchronization, contract updates, invoice recording, and financial transaction monitoring. Users may experience temporary delays in accessing system functions when network connectivity is unstable or unavailable.

Browser Compatibility

The system is designed to operate on modern web browsers such as Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari. Although responsive and standards-compliant, performance may vary when accessed through outdated browser versions that do not fully support current web technologies.

Cloud Server Availability

Since the platform relies on cloud infrastructure for data storage and application hosting, system availability depends on the uptime and reliability of the hosting provider. Scheduled maintenance, server interruptions, or unexpected outages may temporarily affect access to the system.

Data Storage Capacity

The system stores contracts, purchase order confirmations, invoices, logistics records, payment records, and audit logs within a centralized relational database. As operational data accumulates over time, additional storage resources and database optimization may be required to maintain system performance.

Non-Technical Constraints

User Adoption and Training

The operational personnel of Heindrich Trading Co. currently rely on spreadsheet-based workflows and manual documentation processes. Transitioning to a centralized management information system may require user orientation and training to ensure proper utilization of system features and workflows.

Development Timeline

The project follows the Agile Software Development Methodology and is constrained by the Capstone 1 and Capstone 2 academic schedule. Development activities, including requirement gathering, system design, implementation, testing, and deployment, must be completed within the allotted academic timeframe.

Data Confidentiality

Heindrich Trading Co. handles sensitive procurement contracts, supplier pricing information, customer transactions, and financing records. Due to the confidential nature of these business operations, access to certain data must remain restricted to authorized personnel and may limit the amount of real-world financial information available during system development and testing.

Organizational Scope

The proposed system is designed specifically for the internal operational requirements of Heindrich Trading Co. The platform focuses on procurement management, logistics monitoring, invoicing, and payment-to-expense matching. Functions such as full accounting, tax management, regulatory compliance automation, public customer portals, and e-commerce services are outside the scope of the project.

Resource and Budget Limitations

As an academic capstone project, the development team operates under limited financial and technical resources. The system implementation prioritizes cost-effective technologies and open-source development tools while avoiding expensive hardware integrations such as RFID devices, GPS tracking systems, and enterprise-scale ERP solutions.

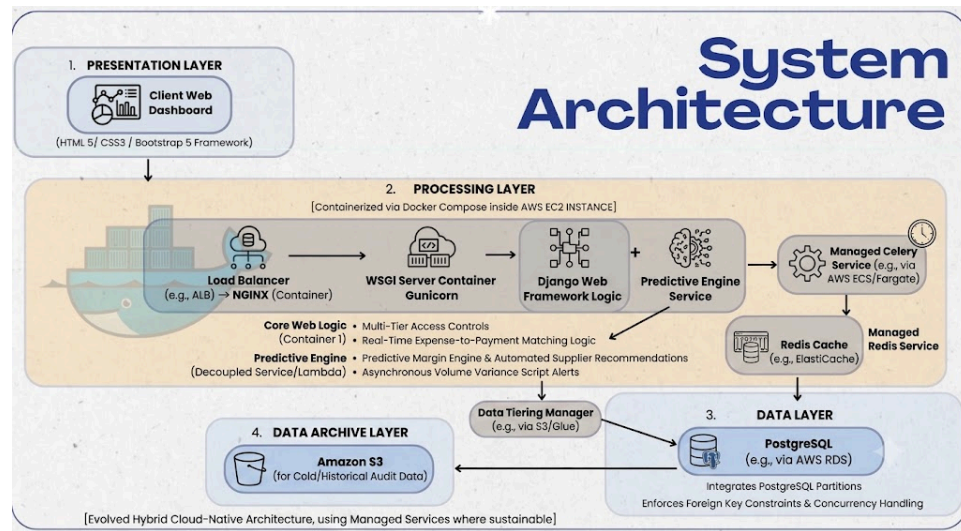


Figure 3.4 – System Architecture

To eliminate the financial burden of always-on server hosting while maintaining a resilient, enterprise-grade runtime environment, the platform transitions from a traditional cloud server model to an Event-Driven, Scale-to-Zero Serverless Architecture. As established in Chapter 2, a primary criticism of modern digital supply chain frameworks is their high operational cost and capital-intensive infrastructure dependencies. This optimized architecture addresses this gap by ensuring that computing resources shrink to absolute zero when the platform is idle, incurring costs only when transactional data is actively processed. The system isolates its infrastructure into three independent operational layers while maintaining a portable Docker container standard to protect future migration pathways back to a dedicated server (e.g., Amazon EC2) if concurrent traffic volumes demand a continuous runtime environment.

Presentation Layer (Static Hosting & CDN)

The client-facing frontend application—constructed using HTML5, CSS3, and the Bootstrap 5 framework—is completely decoupled from the processing layer. Rather than utilizing processing server cycles to serve static interface views, the frontend assets are deployed directly to serverless edge networks such as Vercel or GitHub Pages. These platforms distribute the dashboard assets globally across a Content Delivery Network (CDN). When administrative or operations staff access the platform, the static web environment is instantly delivered to their browser with near-zero latency, eliminating web server performance bottlenecks and keeping the presentation baseline cost at exactly P0.00.

Processing Layer (Event-Driven Cloud Compute)

The backend application environment manages dynamic core business logic, including multi-tier access controls, contract tracking, and real-time payment-to-expense matching. Rather than executing continuously via Unicorn on a dedicated server instance, this layer leverages an asynchronous, serverless compute model: AWS API Gateway: Acts as the secure interface entry point. It captures incoming HTTP requests from the presentation layer, handles routing, and handles traffic validation. It incurs zero costs when idle. AWS Lambda (Django Environment): Upon receiving a request from the API Gateway, AWS Lambda dynamically instantiates a lightweight, temporary container containing the Django application core. Django executes the requested business logic (such as compiling per-customer dashboards or validating data entry), returns the dynamic response, and immediately shuts down. Because AWS provides a generous tier of one million free invocations per month, computing costs remain zero during periods of

company inactivity. AWS SQS (Simple Queue Service) & Background Logic: In alignment with Specific Objective 3, resource-intensive operations—specifically the Predictive Margin Engine and the Asynchronous Volume Variance Tracker—must be calculated independently of the main user thread to avoid application latency. Instead of maintaining an always-on Redis cache broker and Celery worker process 24/7, the application drops message tasks into AWS SQS. SQS acts as a serverless queue that triggers a dedicated AWS Lambda function to compute volumetric shipping shrinkage against standard tolerances or evaluate compounding loan interest clocks on-demand. SQS scales down to zero tasks when idle, preserving the out-of-pocket sustainability of the framework.

Data Layer (Serverless Relational Database)

The persistent storage engine manages the enterprise's relational tables, enforcing data constraints, database-level foreign key validation, and PostgreSQL partitions. Traditional managed databases require continuous memory allocations that are financially unsustainable for zero-revenue operations. To mitigate this constraint, the platform integrates with a serverless PostgreSQL provider such as Neon or Supabase. This cloud architecture completely separates database storage from database compute resources. While contract logs, audit trails, and financial records remain securely preserved on resilient cloud storage, the active database compute engine automatically "sleeps" after a configurable threshold of zero user activity. The moment an AWS Lambda function issues an operational query (such as fetching a specific Purchase Order number), the database compute engine wakes up instantly, executes the transaction, and returns to sleep—ensuring data integrity and security at a flat zero-dollar idle cost baseline.

Table 3.4 – Frontend Framework

Frontend Framework	Language	Performance	Ease of Use	Project Suitability
Bootstrap 5	HTML, CSS, JavaScript	Lightweight	High (Pre-built)	Optimal: Drastically cuts development time
React.js	JavaScript	Virtual DOM (High state-change speed)	Low (Requires custom styling/logic)	Overkill: Unnecessarily complex for current scope
Vue.js	JavaScript	Virtual DOM (High state-change speed)	Low (Requires custom styling/logic)	Sub-optimal: Requires extra architecture setup

Angular	TypeScript	Real DOM (High memory load)	High (Monolithic framework assets)	Unsuitable: Excessive overhead for project scope.
Tailwind CSS	HTML, CSS, JavaScript	Atomic CSS (Smallest file footprint)	Low (0% Unstyled utility classes)	Sub-optimal: Increases UI building time

This criterion assesses component readiness and architectural performance to determine suitability for the Management Information System (MIS). Bootstrap 5 offers a high component readiness level by providing 100% pre-designed UI elements (tables, buttons), whereas Tailwind CSS requires manual, utility-class configuration for every design element, significantly inflating development hours. Architecturally, while React and Vue leverage a Virtual DOM for dynamic UI state-changes and Tailwind utilizes Atomic CSS to minimize style sheets, these paradigms do not align with the system's primary requirement. Because the MIS prioritizes rapid initial page load times over hyper-reactive interface updates, Bootstrap's lightweight CSS/JS bundle is optimized to meet the platform's specific performance needs.

Table 3.5 – Backend Framework

Backend Framework	Language	Security & ORM	Native Admin & Auth Tools	Project Suitability
Django	Python	Native SQLi/XSS protection	Built-in Admin & Auth	Highly Suitable: Built-in RBAC, Auditing & Logs
Laravel	PHP	Eloquent ORM	Partial (Requires separate packages)	Lacks Native Tools: Requires more setup than Django
Express.js	JavaScript	None (Requires 3rd-party modules)	None (Must code Auth from scratch)	High Effort: Requires third-party libraries
Spring Boot	Java	Hibernate ORM	None (Heavy custom configuration)	Excessive: Too heavy for system scope
Flask	Python	None (Requires SQLAlchemy extension)	None (Micro-framework structure)	Inadequate: Lacks built-in MIS features

This assessment evaluates the architectural security features of development frameworks upon initial installation, specifically contrasting Django's "batteries-included" paradigm with minimalist frameworks like Express.js and Flask. Django natively delivers integrated administration dashboards, user authentication systems, and a robust Object-Relational Mapper (ORM) that automatically sanitizes inputs to mitigate SQL injection risks out-of-the-box. Conversely, Express.js and Flask lack native security

infrastructure and database abstraction layers, necessitating manual, line-by-line configuration that introduces potential vulnerability risks. Consequently, Django's native, pre-configured security components provide a more resilient environment for safeguarding the transactional data logs of Heindrich Trading Co.

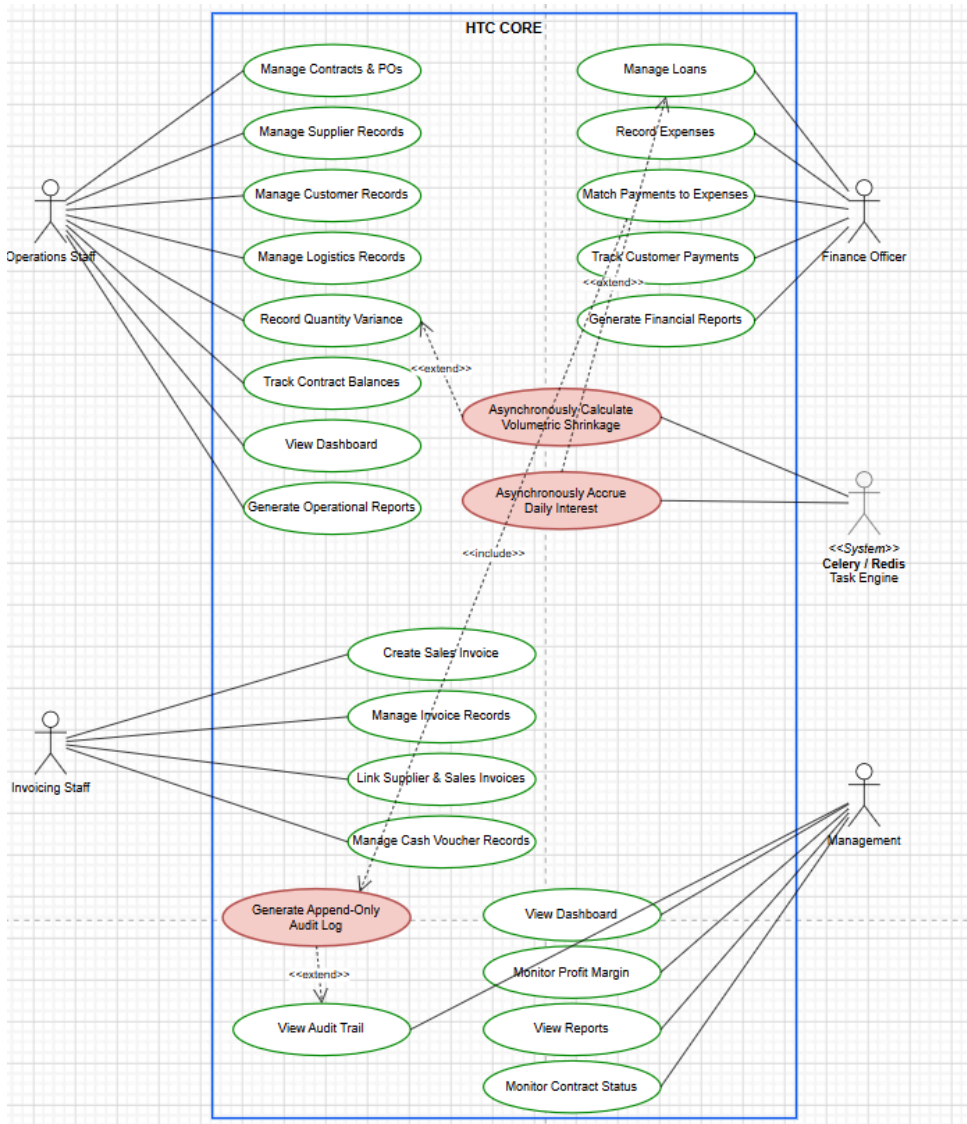
Table 3.6 – Cloud Service Evaluation

Cloud Service	Description	Advantages	Limitations	Suitability for HTC System
Amazon EC2 (AWS)	Virtual cloud server that allows deployment of custom web applications using Django, PostgreSQL, Gunicorn, and NGINX.	Full server control, scalable resources, strong security features, supports custom software stacks, pay-as-you-use pricing.	Requires basic server administration knowledge.	Highly Suitable. Supports the proposed architecture (Django, Gunicorn, NGINX, PostgreSQL), provides secure cloud hosting, and

				allows future system expansion.
Microsoft Azure Virtual Machines	Cloud-based virtual servers for hosting enterprise applications.	Strong integration with Microsoft products and enterprise environments.	Higher complexity and potentially higher costs for small projects.	Suitable but may be excessive for a small-scale commodity trading company.
Google Cloud Compute Engine	Infrastructure-as-a-Service (IaaS) platform for deploying custom applications.	High performance and flexible configurations.	Requires additional configuration and management.	Suitable but provides fewer educational resources and community support compared to AWS.

Heroku	Platform-as-a-S ervice (PaaS) for deploying web applications.	Easy deployment and minimal server management.	Limited control over infrastructure and higher costs as usage increases.	Suitable for prototypes but not ideal for long-term operational deployment.
DigitalOcean Droplets	Simple cloud virtual machines for hosting web applications.	Affordable pricing and easy setup.	Fewer enterprise-level services and scalability options.	Suitable for small projects but less scalable than AWS EC2.

Figure 3.7 – Use Case Diagram

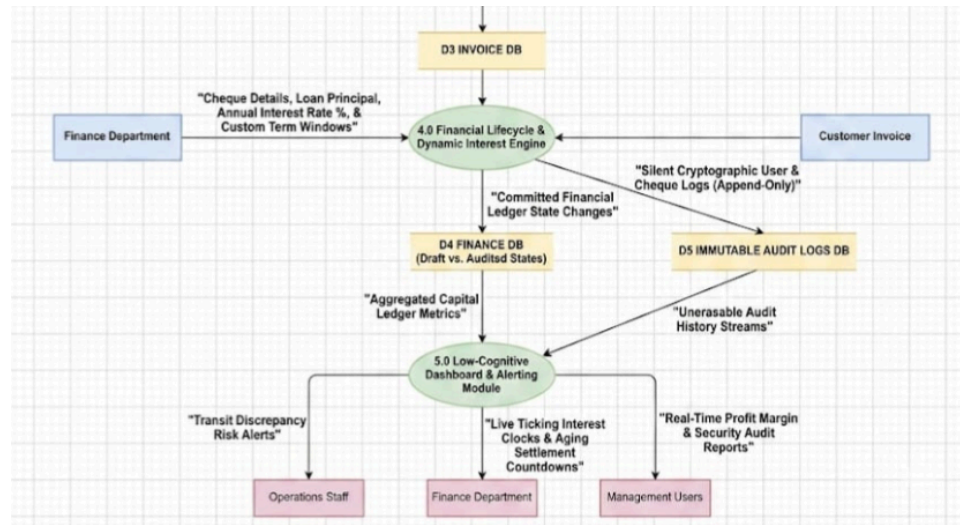


This architectural integration directly bridges human operational workflows with an automated backend task engine to enhance both application performance and ledger data integrity. When the Operations Staff records variance metrics or a Finance Officer manages a loan, the system utilizes an «extend» relationship to hand off heavy

computations to a background worker powered by Celery and Redis. This prevents synchronous blocking of the user interface, allowing volumetric shrinkage tracking and daily interest accruals to execute asynchronously without causing system latency for the user. Simultaneously, critical financial actions—such as matching payments to expenses—utilize an «include» relationship to automatically trigger an immutable, append-only system audit log task. This automated data pipeline ensures that the physical database layers stay synchronized and secure, directly feeding data to the real-time dashboards where management can monitor gross operational margin tracking, review immutable audit trails, and track contractual statuses with complete transparency.

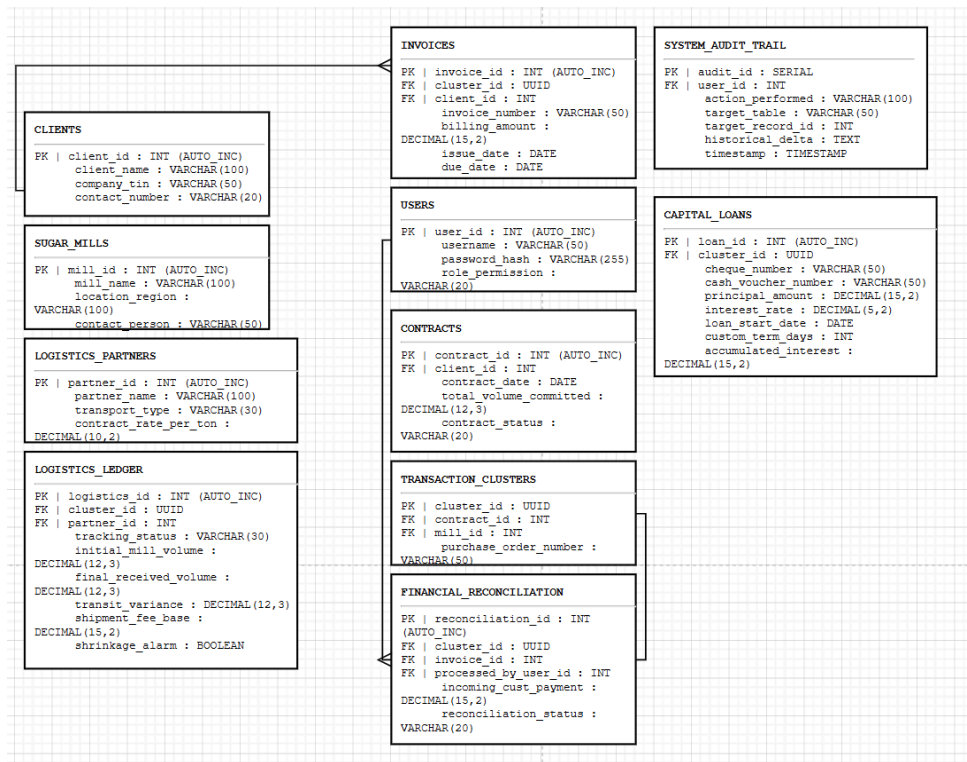
Figure 3.8 – Data Flow Diagram





Data flows sequentially through five core processes to transform raw procurement inputs into actionable system alerts and role-specific metrics. Process 1.0 logs baseline supplier agreements into the Contract Database (D1), which Process 2.0 then pulls to handle asynchronous weight verification and shrinkage matrix generation within the Logistics Database (D2) without delaying the user interface. In parallel, Process 3.0 commits billing records to the Invoice Database (D3). Process 4.0 ingests these invoices and active loan parameters to manage total liabilities within the Finance Database (D4)—which separates tentative drafts from finalized states—while silently exporting cryptographic log streams to the Immutable Audit Logs Database (D5). Finally, Process 5.0 aggregates data from D4 and D5 to deliver tailored operational intelligence: transit risk alerts for Operations Staff, live interest countdowns for the Finance Department, and streamlined gross operational margin overviews for Management, clarifying that these are calculated purely as contract revenue minus procurement and logistical transport costs.

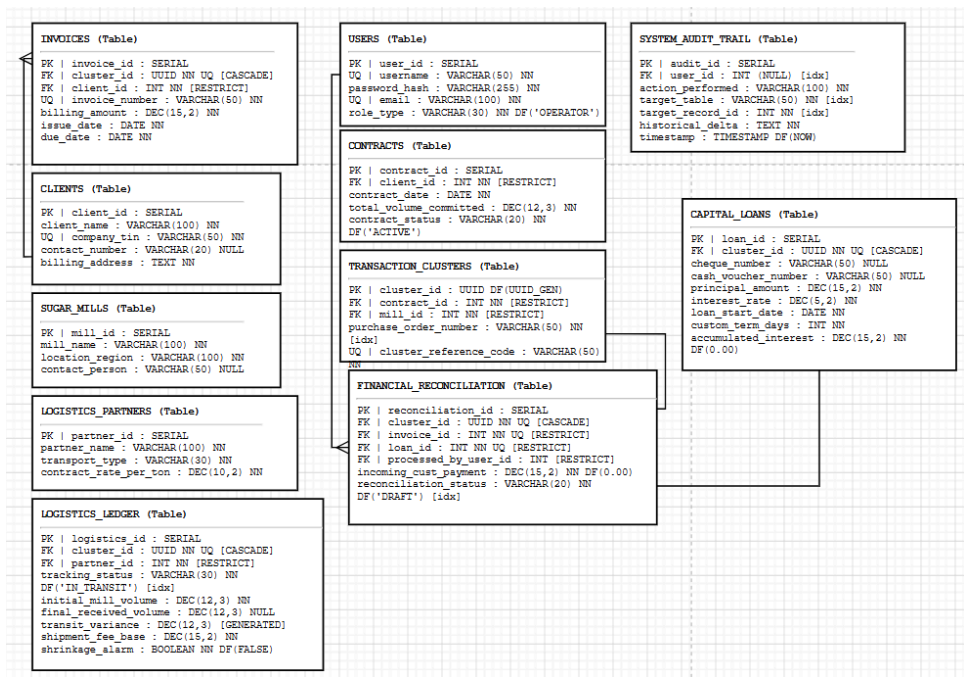
Figure 3.9 – Entity Relationship Diagram (ERD)



This diagram separates master registries from operational data to maintain relational integrity and eliminate redundant data entry. Master tables (CLIENTS, SUGAR_MILLS, and LOGISTICS_PARTNERS) centralize core partner metrics, ensuring uniform TINs, addresses, and freight rates across the system. At the center of the operational core, TRANSACTION_CLUSTERS links customer purchase orders and supplier contracts to all downstream tracking modules. Physical inventory risks are managed via the LOGISTICS_LEDGER through automated transit variance calculation,

while financial exposure is monitored in real time via the CAPITAL_LOANS dynamic interest engine. For compliance and security, INVOICES cross-reference transactions directly to bypass intermediate queries, FINANCIAL_RECONCILIATION enforces strict state-driven approval workflows, and a decoupled SYSTEM_AUDIT_TRAIL captures append-only historical deltas using a scalable polymorphic design.

Figure 4.0 – Database Schema



The physical database schema is structured into distinct functional layers to ensure strict relational integrity, optimize query performance, and eliminate data redundancy. Foundational master registries consisting of CLIENTS, SUGAR_MILLS, and LOGISTICS_PARTNERS centralize core entity profiles, ensuring that uniform billing, geographical data, and contract freight rates remain consistent across all operations. At

the heart of the transactional layer sits the TRANSACTION_CLUSTERS table. Utilizing unique UUID identifiers, this table serves as a centralized hub that binds upstream supplier contracts directly to individual sub-modules via rigid 1:1 constraints, simultaneously mapping real-time cargo tracking details in the LOGISTICS_LEDGER and credit exposure in the CAPITAL_LOANS engine. Financial oversight, system security, and data accountability are managed through state-driven workflows and decoupled logging mechanisms. The FINANCIAL_RECONCILIATION table acts as the final gatekeeper for transaction processing, enforcing sequence control by tracking entries through a strict workflow that separates tentative draft statuses from finalized, audited accounting records. To capture a complete historical narrative of the database without degrading performance, the SYSTEM_AUDIT_TRAIL table implements a polymorphic architecture. By recording generic table-string names and corresponding record IDs rather than relying on standard database foreign keys, this append-only tracking engine captures absolute, unerasable historical deltas across every operational table while ensuring the underlying storage engine remains fully scalable as the application grows.

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Appendices

Appendix A

An interview was conducted on May 31, 2026, with Ms. Jewel Marie A. Perez, Operations Staff of the company, to gather information regarding the organization's operational processes, financial monitoring, inventory management, logistics coordination, and existing system challenges. The interview revealed that the company primarily operates as a trader of molasses, sourcing products from local sugar mills and occasionally through importation. The company's major clients include San Miguel, Tanduay, Emperador, Absolut, and other distillery companies.

The company's transaction process begins with client negotiations handled by the operations personnel. Once a deal is finalized, products are purchased from suppliers and delivered to customers. Sales invoices are then prepared and submitted to clients for payment processing. The company also manages various logistics services, including barging and tracking services, which are monitored throughout the transportation process.

A significant portion of the company's operations relies on Microsoft Excel spreadsheets. These spreadsheets are used to record purchase and selling prices, customer information, supplier details, transportation services, invoice numbers, quantities delivered and received, and profit margins. The Operations Staff consolidates and monitors all transaction data to ensure accurate tracking of expenses, revenues, and financial performance.

The interview also highlighted several operational challenges. The current system lacks centralized data management, making information accessible only to specific personnel. Updates must be manually communicated and recorded, resulting in inefficiencies and potential errors. Financial monitoring is also challenging because customer payments, supplier expenses, loans, and service costs must be manually matched and tracked. Additionally, employees rely heavily on paper-based documents, leading to duplicate printing, difficulty locating records, and inefficient document management.

According to the participant, the company would benefit from a user-friendly system that centralizes operational, financial, inventory, and logistics information. The proposed system should provide real-time access to transaction records, automate data organization, simplify monitoring of contracts and deliveries, support financial tracking, and present information through dashboards and summarized reports. Such a system would improve operational efficiency, enhance data accessibility, reduce manual work, and support better decision-making within the organization


Jewel Marie A. Perez
Operations Staff

Appendix B

Attached hereto is the 2026 consolidated Excel sheet for HTC, as submitted by the aforementioned employee. The company reserved the right to redact any information it deems necessary to protect proprietary data and trade secrets.

HEINRICH TRADING CORPORATION 2026												
SI	INV DATE	BARGE	SOURCE	PURCHASED PRICE	LOGISTIC COST TRUCKING	LOGISTICS COST BARGING	CUSTOMER	QUANTITY DELIVERED	QUANTITY RECEIVED	DISCREPANCY	SELLING PRICE	AMOUNT RECEIVABLES
3185	1/13/2026	FRONTIER	LOPEZ			2,095.00	ADN	2001.63	1998.22	3.41		18,278,388.00
3187	1/13/2026	TRUCKING	SEITI PHILUA	165.00			PROGREN		1545.3			15,000,000.00
3188	1/13/2026	MODESTO	CASA SEITI PHILUA	165.00	2,340.00		PROGREN	698.558	899.663	6.23		9,176,960.40
3189	1/14/2026	YANIGTZE	CASA SEITI PHILUA	165.00	2,340.00		PROGREN	1134.991	1202.792	-68.95		14,256,866.40
3190	1/18/2026	TRUCKING	HTE CANA	175.00		2,085.00	ADN	1620.10	1627.86	7.75		15,963,132.00
3196	1/20/2026	TRUCKING	LOPEZ			2,085.00	ADN	1163.90	1138.68	25.22		13,895,000.00
3199	1/26/2026	NIGEL	LOPEZ			2,085.00	ADN	1929.810	1920.87	8.94		23,434,416.00
3210	1/26/2026	TRUCKING	HTE TAN	175.00			ADN	903.2	902.58	0.62		13,011,620.00
3220	2/03/2026	TRUCKING	HTE CANA	175.00			ADN	892.12	893.09	-0.97		19,000,000.00
3231	2/11/2026	FS2	BUSCO	165.00	2,340.00		OSAM	2411.304	2387.95	43.35		8507
3234	2/17/2026	ESADRA	BUSCO		1,480.00		OSAM	2207.38	2112.065	95.32		19,275,180.00
3235	2/17/2026	PHSY	BUSCO		2,340.00		HTE	2412.978	2313.94	100.05		18,464,840.00
3277	3/1/2026	CHARTY	BUSCO		1,480.00		OSAM	1791.612	1760.435	41.18		16,464,840.00
3278	3/11/2026	HARTY	AARE		2,600.00		OSAM	2178.58	2167.885	10.70		16,464,840.00
3279	3/11/2026	YANIGTZE	CASA	495.00	1,745.00		PROGREN	1478.96	1485.38	6.42		15,160,000.00
3280	3/11/2026	1ST KIRSTEN MARIE	CASA	495.00	2,200.00		PROGREN	805.56				REV OF CASA 1500, BL 2
							PROGREN	120.95				
3281	3/11/2026	CHARTY	BUSCO		2,200.00		PROGREN	1,740.72	2441.48	-21.15		14,023,426.00
3292	3/14/2026	2ND VIOLETTE	BUSCO		2,340.00		BDI	1,762.19	1779.03	16.80		18,475,120.00
3293	3/14/2026	BARTS	BUSCO				PROGREN	1,469.95	1431.323	0.89		35,769,742.40
3295	3/27/2026	EMILY	BUSCO				OSAM	1100	1331.41	0.48		15,560,750.20
3297	3/27/2026	ESADRA	BUSCO		2,340.00		OSAM	2205.38	2230.75	25.44		17,618,000.00
3298	3/27/2026	ESADRA	BUSCO		3,500.00		BDI	1846.526	1847.164	0.63		23,316,712.00
3297	3/31/2026	HARMONY	BUSCO		3,080.00		BDI	2500.45	2495.92	0.53		28,663,328.00
3317	4/30/2026	2ND KIRSTEN MARIE	CASA		750.00	1,200.00	EMPERADOR	2439.03	2439.03	-0.09		19,464,400.00
					640.00			659.285	277.85			REV OF CASA 2500 THE 1ST (2439.03) + 609.2
3318	4/10/2026	2ND KIRSTEN RAE	CASA		480.00	2,200.00	EMPERADOR	600	1755.4	25.40		170.40 MT (BY 1000) REV. OF CASA 21
3354	4/10/2026	TRUCKING	HTE		600.00		EMPERADOR	1900	370			17,000,000.00
3355	4/10/2026	TRUCKING	CASA		205.00		EMPERADOR	652.8	652.31	-0.49		
3357	4/13/2026	3RD VIOLETTE	BUSCO			2,680.00	BDI	3452.911	3453.27	0.37		26,006,100.00
3358	4/20/2026	HARMONY	BUSCO		2,675.78		BDI	2137.995	2138.911	-0.91		26,006,100.00
3359	4/22/2026	1ST APRIL SHEIN	BUSCO		2,480.00		BDI	2800.008	2801.383	1.38		19,327,799.80
3361	4/24/2026	HOPE	BUSCO		3,200.00		OSAM	1906.16	1917.59	-11.43		26,006,100.00
3362	4/27/2026	1ST APOSTLE	CAB		2,800.00		BDI	410	2136.97	0.83		26,006,100.00
3366	4/29/2026	4TH VIOLETTE	BUSCO		2,981.00		BDI	1725.361	3449.435	-0.73		16,998,262.20
3367	5/4/2026	FS2	BUSCO		3,500.00		BDI	2448.135	2450.15	1.01		40,000,000.00
3372	5/6/2026	2ND APRIL SHEIN	BUSCO		2,981.00		BDI	2796.593	2797.341	0.75		24,044,278.40
3373	5/6/2026	HOPE	BUSCO		2,800.00		BDI	1906.194	1906.771	0.58		26,006,100.00
3374	5/6/2026	AMAZON	CAB		2,940.00		EMPERADOR	1592.11	1582.69	-9.42		16,844,613.00
3384	5/13/2026	YANIGTZE	CAB		2,940.00		EMPERADOR	1411.629	1424.108	11.65		16,844,613.00
3376	5/16/2026	TRUCKING	YANIGTZE	255.00			BDI	1556.46	1546.46			16,844,613.00
3385	5/13/2026	3RD KIRSTEN RAE	CASA	6,591.00	925.00	2,330.00	EMPERADOR	2100.335	1745.543	2.34	11,500	26,221,661.20
									357.13			1356.25 MT (BY 1000) REV. OF CASA 21
3388	5/15/2026	2ND APOSTLE	CAB		2,800.00		BDI	2112.903	2113.584	0.68		16,460,712.00
3389	5/15/2026	FS2	BUSCO		2,906.23		HTE	2451.778	2456.891	5.11		
3389	5/16/2026	3TH VIOLETTE	BUSCO		3,450.00		BDI	1931.73	1931.528	-1.80		16,460,712.00
3394	5/21/2026	HOPE	URC-PASSI		2,800.00		BDI	1900.97	1900.969	-0.01		26,006,100.00
3396	5/29/2026	3RD APOSTLE	URC-PASSI		2,332.00		BDI	2800.11	2799.818	-0.29		44,770,000.00
3397	5/29/2026	3RD APOSTLE	URC - SONEDCO			2,500.00	BDI	2100	2109.226	9.22		14,460,000.00
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